Restricting an acquired subgroup inventory to an arbitrary subgroup

L054 reuse component

This elementary extension of the accepted normal-subgroup restriction method uses actual embedded subgroups. It is not a new finite-group classification. Conjugation is on the right: $H^g = g^{-1}Hg$, hence $(H^g)^k = H^{gk}$.

Theorem 1. *Let G be finite, $K \leq G$, and $\mathcal{R}$ a complete covering list of subgroup G-conjugacy classes. Let T represent the left cosets tK. For every $H \in \mathcal{R}$ and every $t \in T$, form H^t and keep it if and only if $H^t \leq K$. Fuse these candidates by actual K-conjugacy. The surviving list represents every subgroup K-conjugacy class exactly once. Input repetitions are harmless.*

Proof. Given $L \leq K$, completeness supplies $H \in \mathcal{R}$ and $g \in G$ with $L = H^g$. Write $g = tk$, where $t \in T$ and $k \in K$. Then $L = (H^t)^k$, and $H^t = L^{k^{-1}} \leq K$. Thus a candidate in the K-class of L survives the containment test. Conversely every retained candidate is a subgroup of K. Removing a candidate only after positive K-conjugacy to an earlier survivor preserves coverage. Testing all earlier survivors makes the final list irredundant. Input repetitions only repeat candidate classes and do not change this argument. □

Neither normality nor an initial test $H \leq K$ is permissible as an unstated hypothesis. A chosen ambient representative may lie outside K even though its class meets K. The code therefore tests containment only *after* conjugating each ambient representative. It records the exact origin (H, t) and the final class of every admitted candidate.

GAP's `RightTransversal(G,K)` represents right cosets Kt. Inverting these representatives gives the needed left transversal: inversion bijects Kt with $t^{-1}K$. The lower-level interface checks $|T| = [G : K]$, each $t \in G$, and $t_i^{-1}t_j \notin K$ for distinct entries. These checks establish left-coset coverage. They do not establish completeness of $\mathcal{R}$, which must be acquired from its original certificate. Input containment and the presence of G and of the final top K are additional sanity checks, not substitutes for that premise.

Corollary 2. *The acquired robust local-weight method can compute the complete weighted maximal-incidence table from this restricted inventory without enumerating all subgroups of K again.*

Proof. For each retained parent H, acquire a complete cover of its maximal-subgroup classes, using checked faithful degree reduction if desired. Remove local conjugacy repetitions, and transport the actual subgroups back into H. The inventory theorem places each in exactly one K-class. A local class represented by M contains $[H : N_H(M)]$ actual maximal subgroups. Sum these indices whenever several local classes fuse to the same ambient child class. This counts every actual maximal inclusion exactly once. Writing $c_H = [K : N_K(H)]$, double counting incidences gives reverse weight $c_H a_{HL}/c_L$. No cover relation in the conjugacy-class poset is used. This is the already accepted robust weight argument; the present new kernel supplies only its complete inventory input. Ambient G-edge weights cannot in general be reused unchanged after restricting to a nonnormal K. □

Literal controls and strict countercontrols. The finite controls construct all actual subgroups of six tiny permutation groups, through order24, by repeated literal multiplication closure. They then restrict independently conjugated complete ambient class lists to *every* actual subgroup, both with and without input repetitions. Expected K-classes are formed by literal conjugation over all elements of K, separately from the kernel's subgroup-conjugacy decision. The tests check both directions, exact origins, candidate fusion and class lengths.

A nonnormal S_3 fixture chooses $K = \langle (1\,2) \rangle$ and the ambient representative $H = \langle (1\,3) \rangle$. It constructs a genuine right transversal with $H^t \neq K$ for every chosen representative. Using these representatives without inversion misses the target top; inverting them recovers it. Filtering the chosen ambient list by $H \leq K$ before transport also misses the top. Thus both tempting incorrect variants are rejected by actual group examples, not merely by differently written code.

Acquired inputs and scope. Complete ambient inventories and local maximal covers, faithful transports, and exact subgroup/conjugacy/normalizer operations retain their original GAP software and classification premises. The primitive operations are not reproved here. Intended applications are the index3 nonnormal subgroup in the already acquired $P\Gamma U_3(5)$ inventory and small-index subgroups of the acquired full automorphism model of $\mathrm{PSL}_3(4)$, once the required complete ambient inventory is available. Each target still needs its actual named-model identification and source join. This method packet runs no atlas target and grants no numerical, structural-label, whole-source, question, formalization or measurement credit.